

Due: Sep. 28, 2026 @ 11:59 p.m.

General notes to keep in mind:

- ▶ All deliverables for the assignment must be submitted as a **single ZIP file per group** via the Brightspace D2L [course shell](#). Submissions containing multiple ZIP files per group or those with a file that is not in the ZIP format will NOT be graded.
- ▶ The code submitted must be written purely using the [Python programming language](#) and it should execute within the [Python 3.11.9 interpreter](#) running on the Windows operating system (version 10 or above). The submitted code should NOT require external python modules other than [matplotlib 3.10.8](#), [pandas 2.3.3](#) and their dependencies.
- ▶ It is your responsibility to ensure that the submitted code executes. If the grader is unable to execute your code and/or your code does NOT adhere to the submission requirements, your code may not be graded. Make sure to run all cells in order before submitting the notebook, as the figures must be reproducible by running all cells top-to-bottom.
- ▶ The written responses (when required) to the questions in the assignments should be included as a separate cell in the python notebook with the solution.

Plastic Waste on the World and Recommended Hours of Sleep

Instructions for this assignment

There are many levels of customization that can be accomplished with Matplotlib. You are free to use all parts of the Matplotlib API. Some useful information for this assignment include:

- The borders around the frame of the plot are called ‘spines’.
- The colours needed for this assignment are ‘#051B2C’ and ‘#4789BE’.

Data

The datasets `river-plastics.csv` and `sleep.csv` are included within the same .zip file.

The first dataset contains information about plastic waste on a per country basis. Provided is the total mismanaged plastic waste (MPW) given in metric tons per year, and the MPW that is emitted into the oceans (ME), also in metric tons per year. Geographic details of each country are provided, such as its area (in km²), length of coast lines (km), and annual rainfalls (mm / year).

The second dataset contains information on the daily number of hours of sleep that are recommended, appropriate, and not recommended by various age groups. This data was compiled by the National Sleep Foundation in the USA from a panel comprised of members and sleep experts from various American medical associations.

The data comes from the following articles:

- *More than 1000 rivers account for 80% of global riverine plastic emissions into the ocean*, Science, <https://www.science.org/doi/10.1126/sciadv.aaz5803>
- *Journey into sleep: An illustrated tour of why scientists are finding that sleep matters more than we know*, Reuters, <https://www.reuters.com/graphics/HEALTH-SLEEP/mopakyjmnpa/>
- *National Sleep Foundation’s sleep time duration recommendations: methodology and results summary*, Sleep Health, <https://doi.org/10.1016/j.sleh.2014.12.010>
- *Plastic Pollution*, Our World in Data, <https://ourworldindata.org/plastic-pollution>

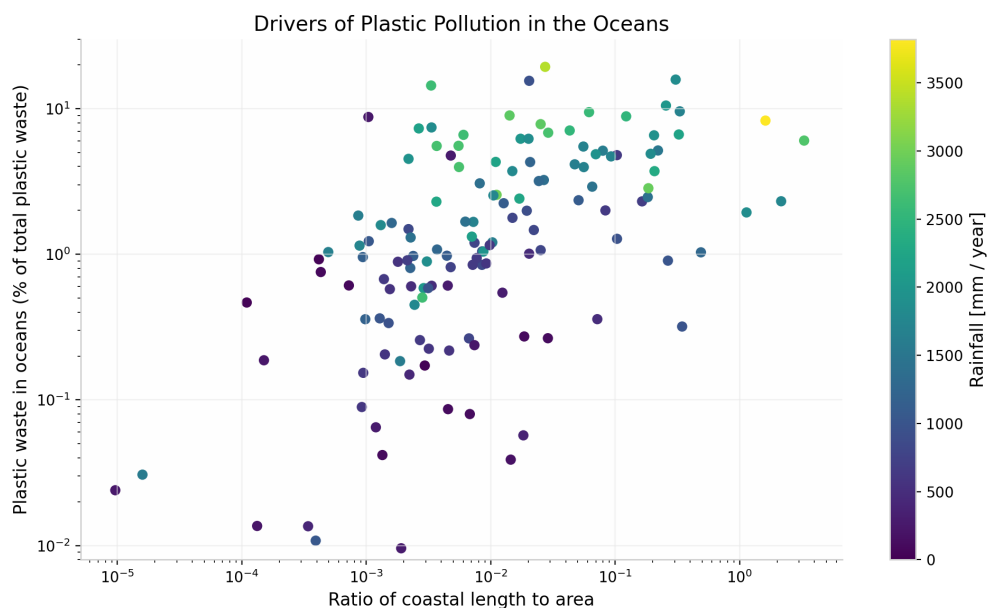
Question 1 - Scatter Plot - Waste by Country (based on coastal line, area and rainfall) [45 marks]

Approximately 1–2 million tons of plastic waste enter the oceans annually, and rivers bring a significant portion of this plastic waste to the ocean.

The plot below shows how a country's coastal length, area and rainfall affect its percentage of plastic waste that is emitted to the oceans. Each point represents a country that has a coast line. Countries with a higher ratio of coastal length to their land area, and that receive higher annual rainfalls, tend to have a larger percentage of their mismanaged plastic waste end up in the oceans.

Using the `river-plastics.csv`, re-create the plot below using Matplotlib. Ensure that all elements are replicated, such as,

- the x-axis is the ratio of coastal length to area,
- the y-axis is the percentage of plastic waste that is emitted into the oceans,
- the provided `Ratio ME/MPW` column may be used to verify your calculation,
- points are coloured according to the country's annual rainfall,
- a colourbar is added to the figure with the default colour map ('viridis') and the correct label (hint: `fig.colorbar()`),
- the range of the colourbar is correct,
- the x- and y-axes use log scale, and
- the axes labels and plot title are present (Title: Drivers of Plastic Pollution in the Oceans, x-label: Ratio of coastal length to area; y-label: Plastic waste in oceans (% of total plastic waste); colourbar: Rainfall [mm / year]).

**Question 2 - Connected Dot Plot - How much should you be sleeping? [55 marks]**

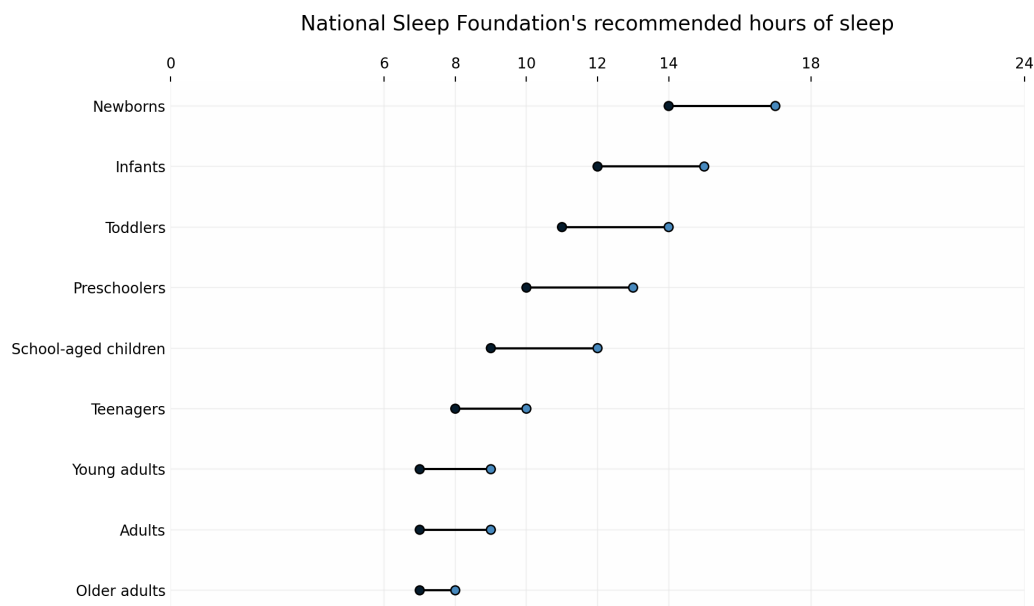
A connected dot plot shows quantitative values for different categories of items, with each item containing a secondary categorical breakdown. Each dot represents a data point, and the dots are connected by a line. This helps enable comparison of values within a single item by displaying the difference (delta) between data points, in addition to comparison of values across multiple items.

A connected dot plot is useful for representing, for example, high and low values or targetted and realized values, though more than two sub-categories is possible as well.

Re-create the plot below using Matplotlib. All elements should be replicated as shown. In particular, ensure that,

- the dots are coloured correctly,
- the category labels along the y-axis are ordered and positioned correctly,
- the x-axis tick marks and labels are correct,
- the x-axis tick marks and labels are above the plot (instead of below),
- the spines along the left, bottom and right side are removed, and
- the title is present.

Also, make sure to answer the following question in an separate cell of the notebook: *Describe how the recommended sleep range changes across the lifespan, and why a connected dot plot communicates this better than two bars per group.*



Note on grading and overall tips

The grading for Question 1 and Question 2 will be based on the appropriateness of the submitted code and the written responses when requested. Submissions should include an attribution section indicating any sources of material, ideas or contribution of others to the submission. Avoid academic penalties by properly attributing any contribution to your submission by others, including internet sources and classmates. This will also help distinguish what elements of the submission are original. You may not receive full credit if your original elements are insufficient, but you can avoid penalties for plagiarism or copying if you acknowledge your sources.